STM Exercise 3

- Creep
- Fatigue

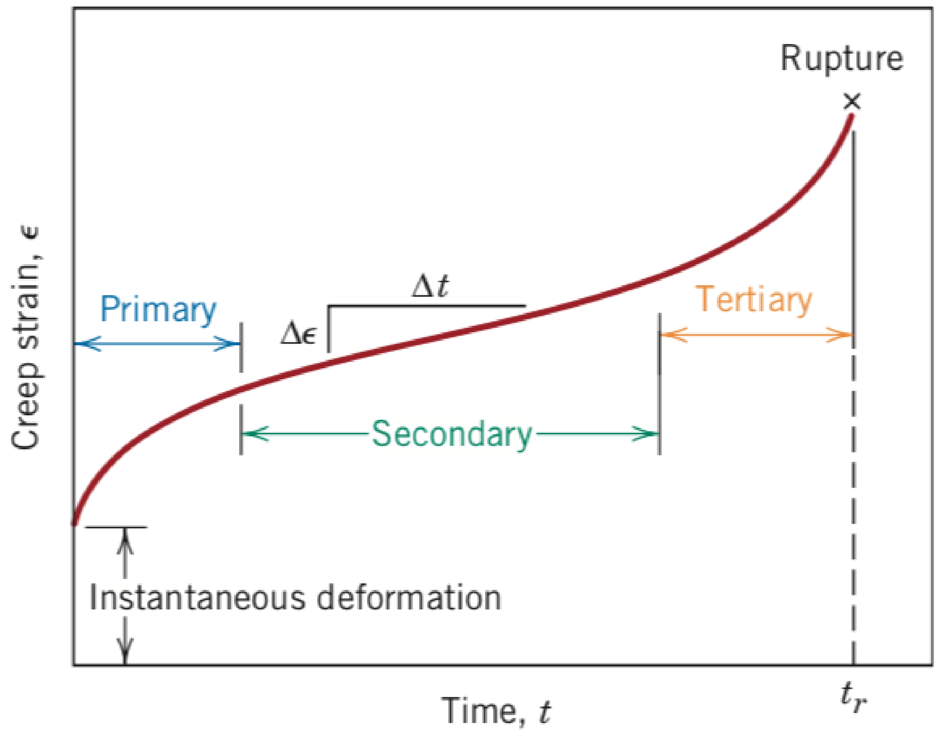
CREEP

Creep test: a constant load is applied at a given T, observing a strain versus time where rupture occurs.

Constant load in the elastic field can give to a plastic deformation, progressive to fracture.

1. Thermally activated process (High T Process)

Metals **T>0,3-0,4 $T_{melting}$**
Ceramics **T> 0,6-0,7 $T_{melting}$**
Amorphous materials **T>T_{glass}**



Ex.1 Draw tensile and creep curves of a given metal with the following microstructures and Briefly explain the differences: mono-crystalline, polycrystalline with large grains, polycrystalline with small grains.

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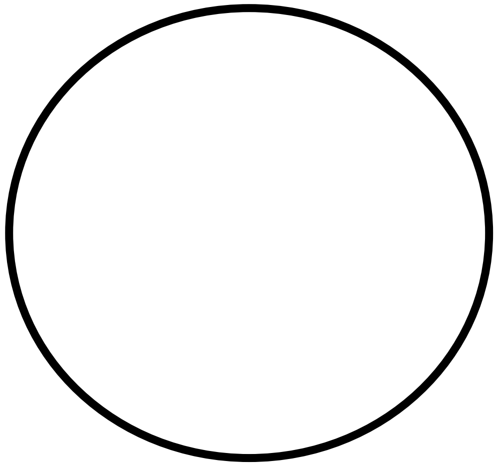
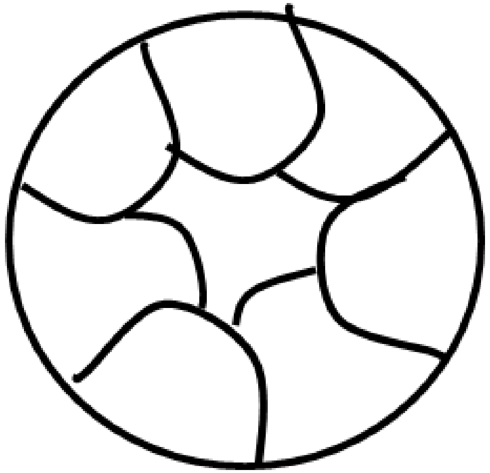
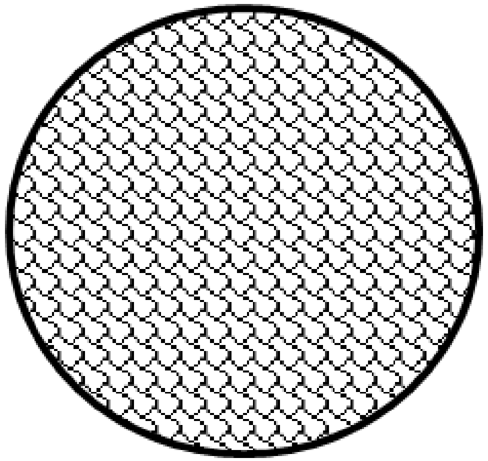


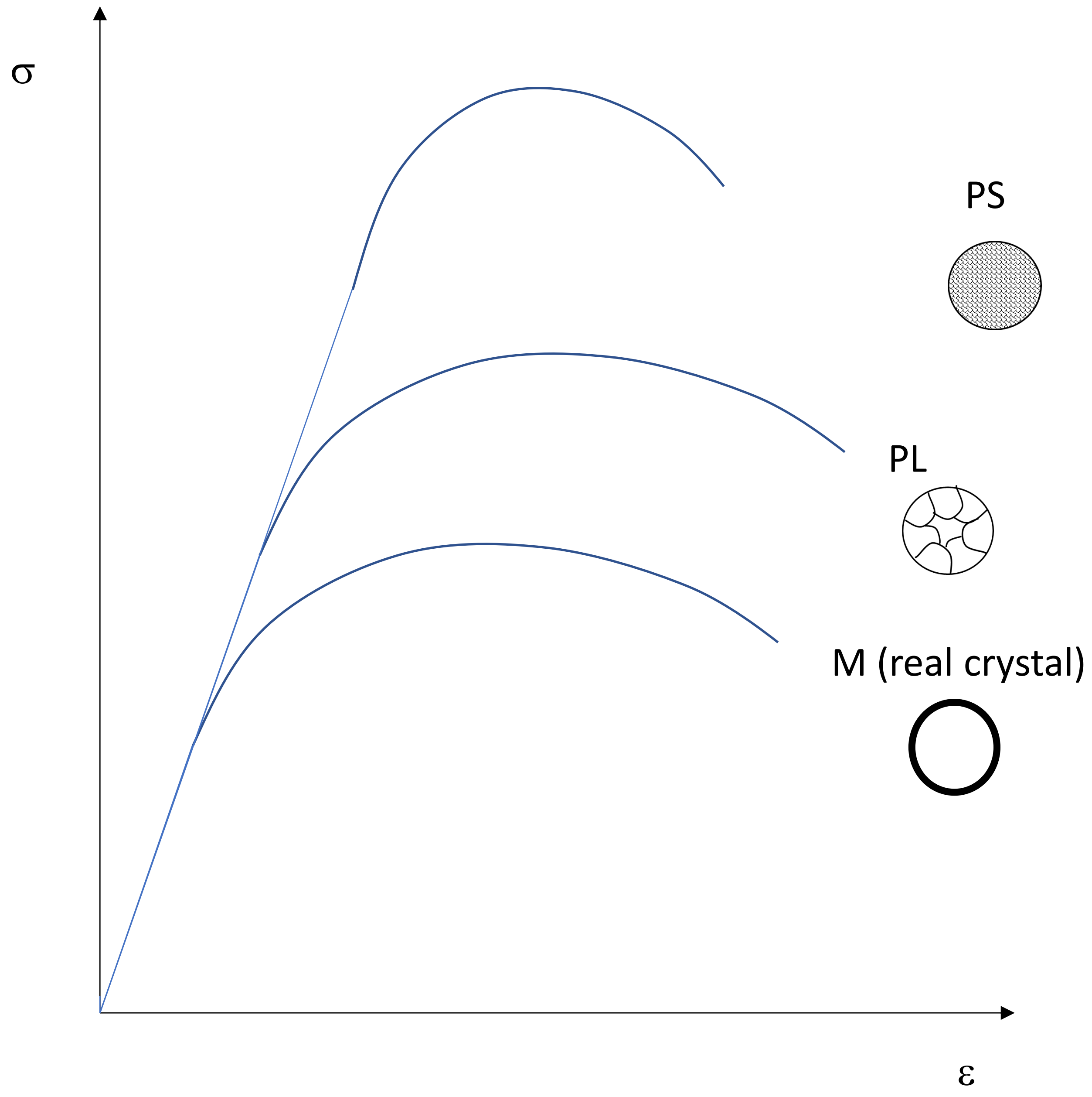
Monocrystalline (M)



Polycrystalline with Large grains (PL)



Polycrystalline with Small grains (PS)



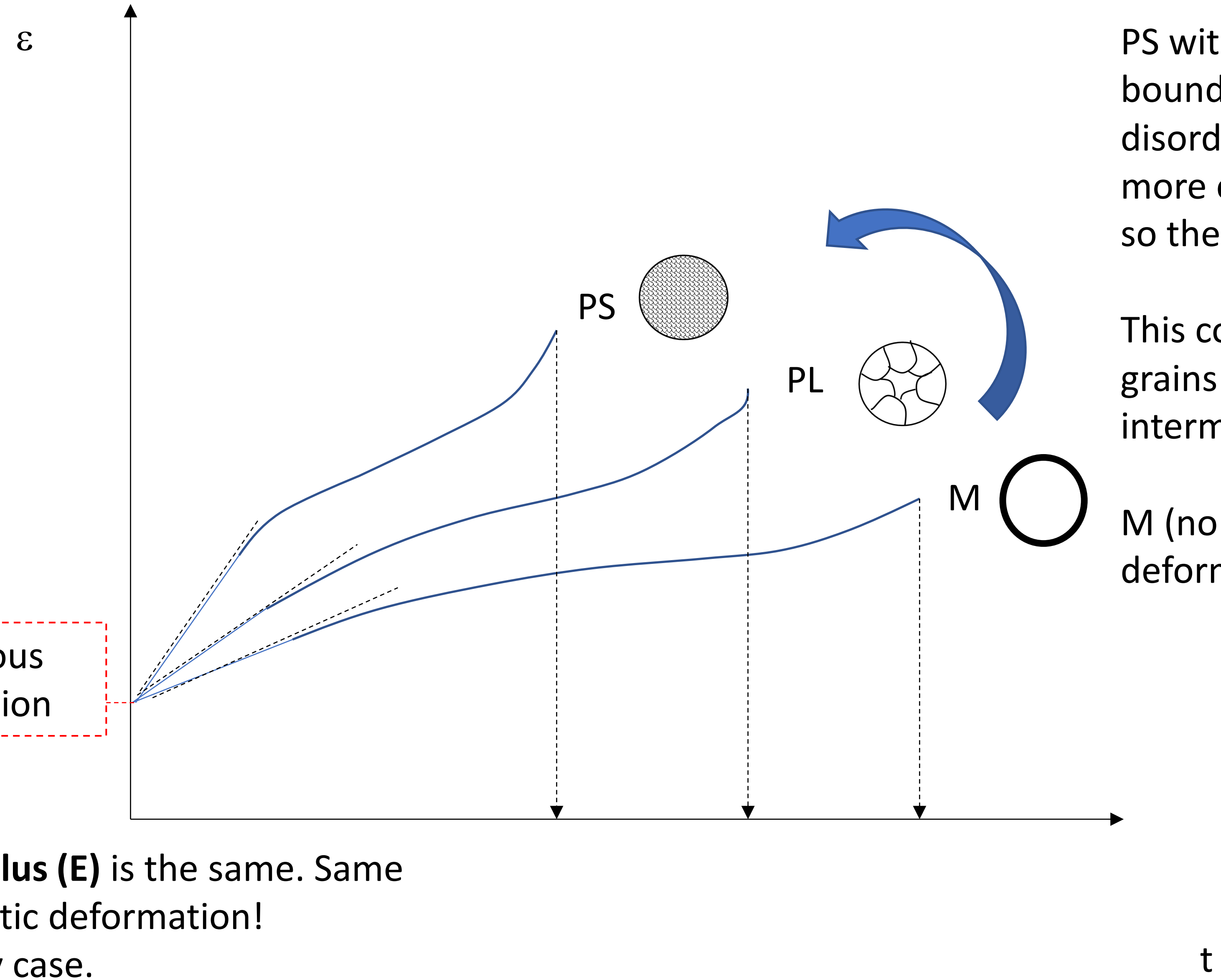
Monocrystal (M real):
→no reinforcing mechanisms →lowest yield strength

Policrystalline large grains (PL):
→few grain boundaries, poor reinforcing mechanism, intermediate yield strength

Polycrystalline small grains (PS): many grain boundaries, strong obstacle to dislocation motion, highest yield strength

The elastic modulus (E)is the same because the metal is the same

Even though the grain size changes, the atomic bonding remains the same, because the chemical composition is identical



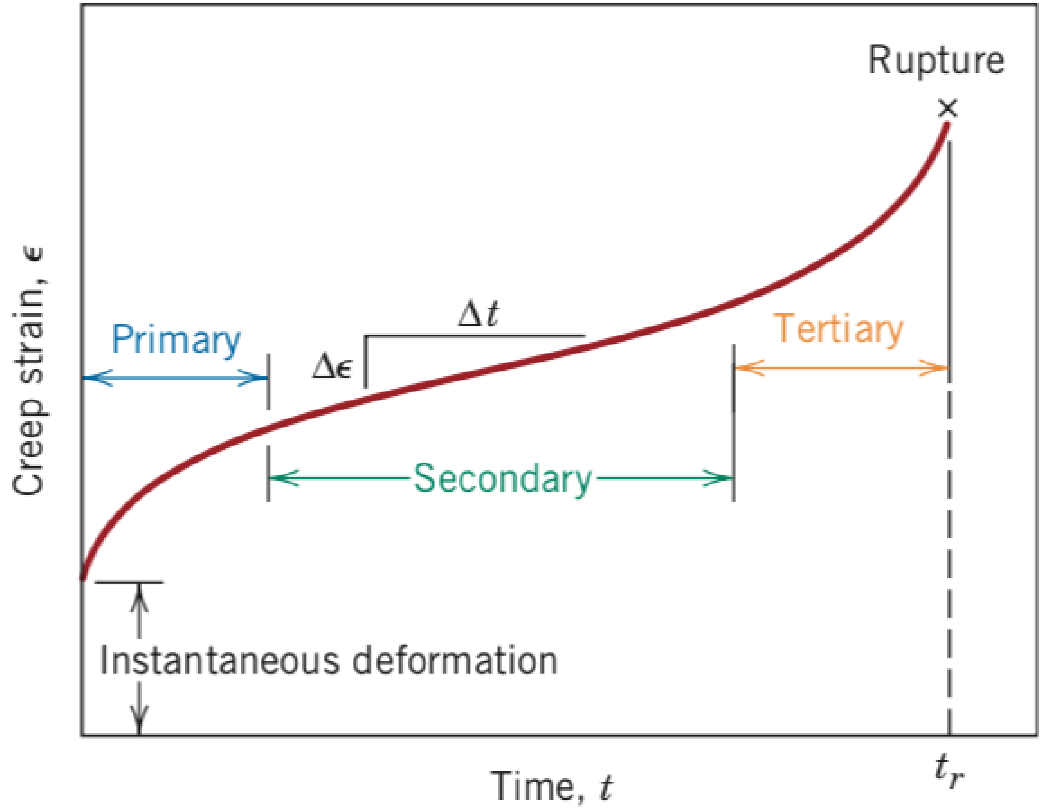
PS with small grains→ many grain boundaries, which are more "open" and disordered regions→Atoms can diffuse more easily along these boundaries→ so the material deforms faster

This compared to one with PL large grains→fewer boundaries → intermediate rate of deformation

M (no grain boundaries) → more slowly deformation at creep and fracture.

The elastic modulus (E) is the same. Same istantaneous elastic deformation!
It's a metal in any case.

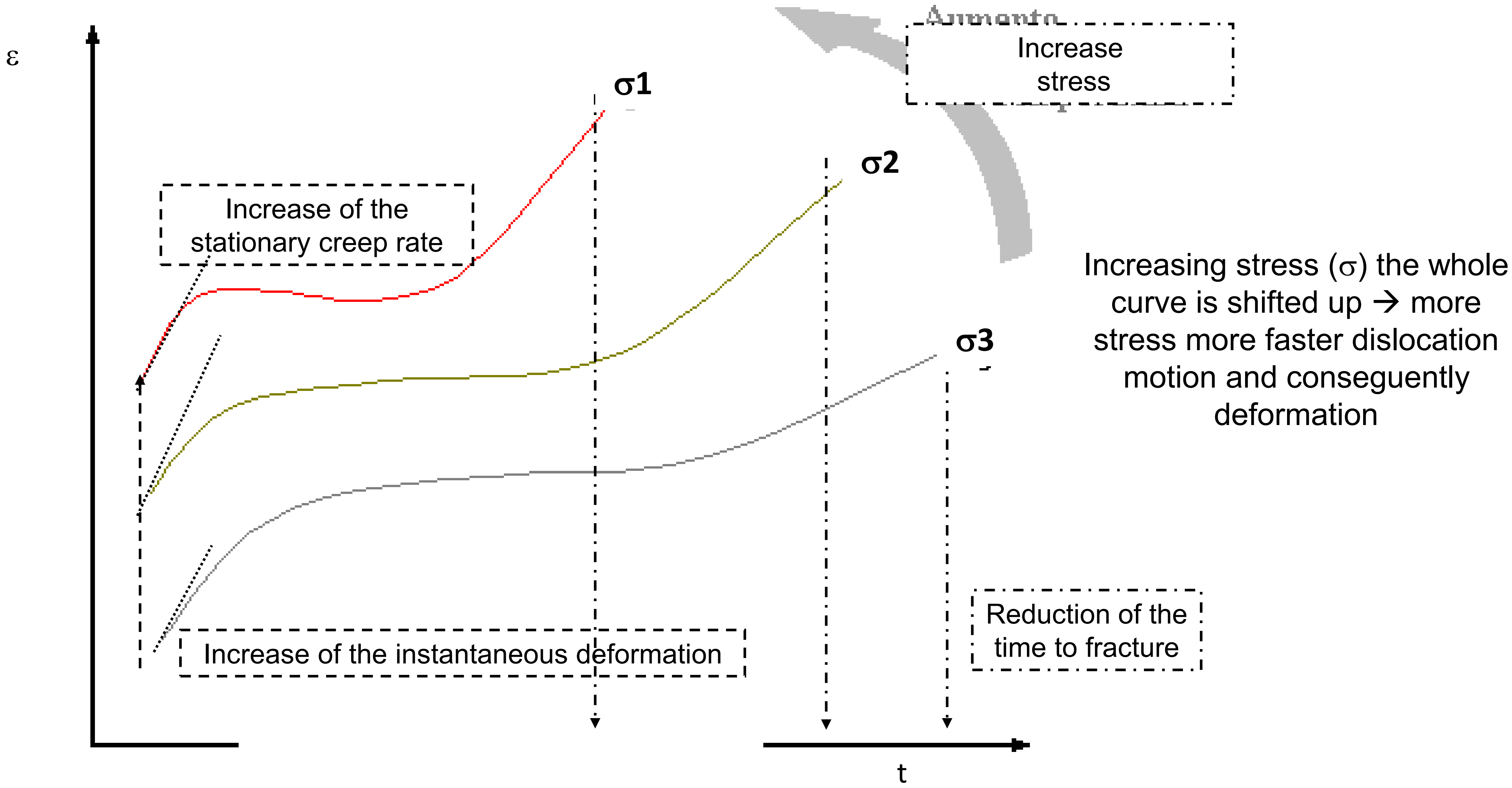
Ex.2 Draw the creep curve of materials and briefly discuss the three phases of creep in term of materials micro-structural modification. Draw the creep curves of the same material at $\square_1 > \square_2$ and at T1>T2. Explain differences.

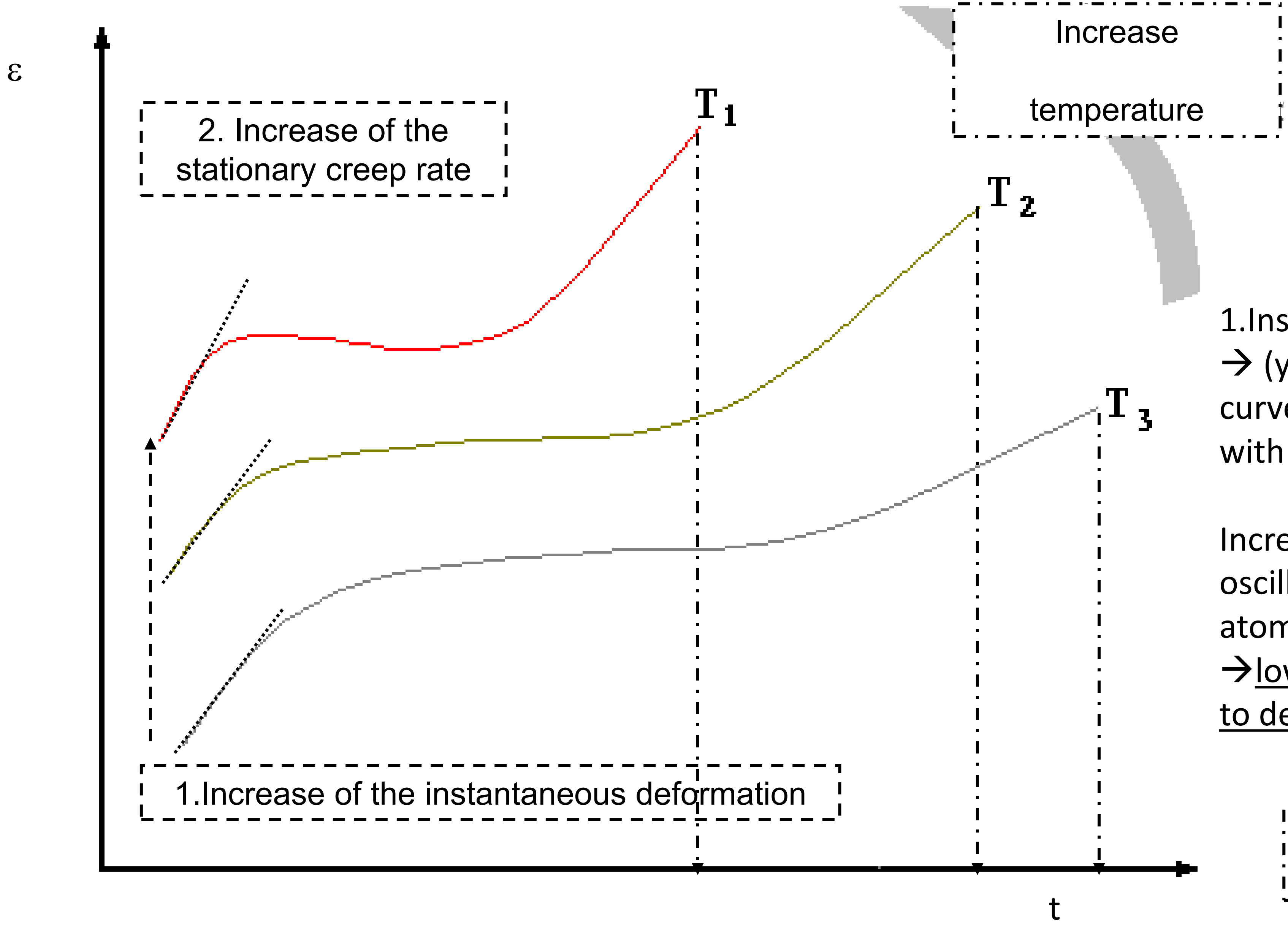


During creep, work hardening initially higher due to dislocations accumulate. Creep rate decrease (Primary creep)

In the steady-state stage, work hardening is balanced by recovery processes →resulting in a constant creep rate. (secondary creep)

In the tertiary stage, microcavities and macro defects at grain boundary dominate. (tertiary creep)


ε
σ1
σ2
σ3
Increase stress
Increase of the stationary creep rate
Increasing stress (σ) the whole curve is shifted up → more stress more faster dislocation motion and conseguently deformation
Reduction of the time to fracture
Increase of the instantaneous deformation
t



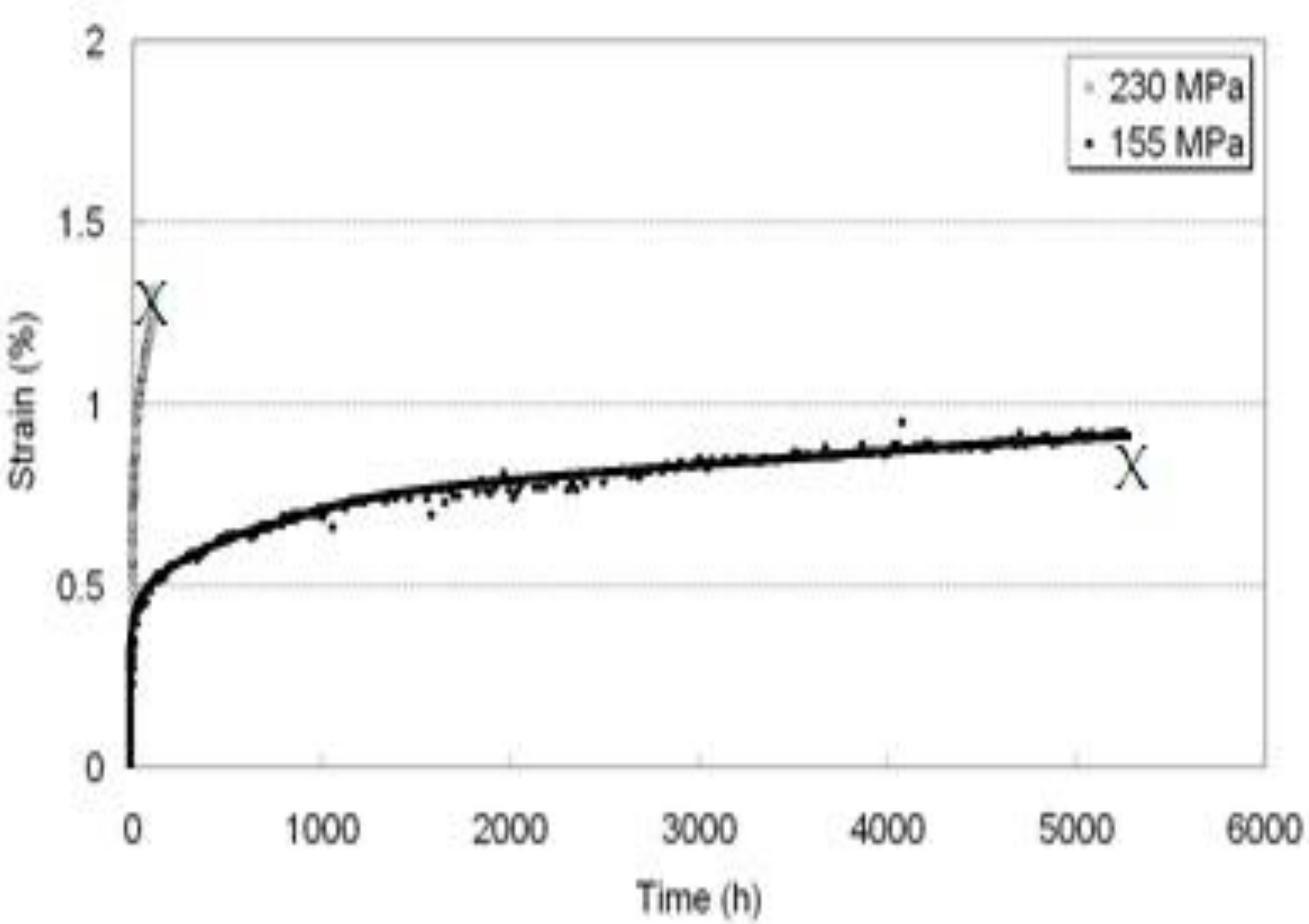
2. Increase stationary creep rate: Dislocations move more easily

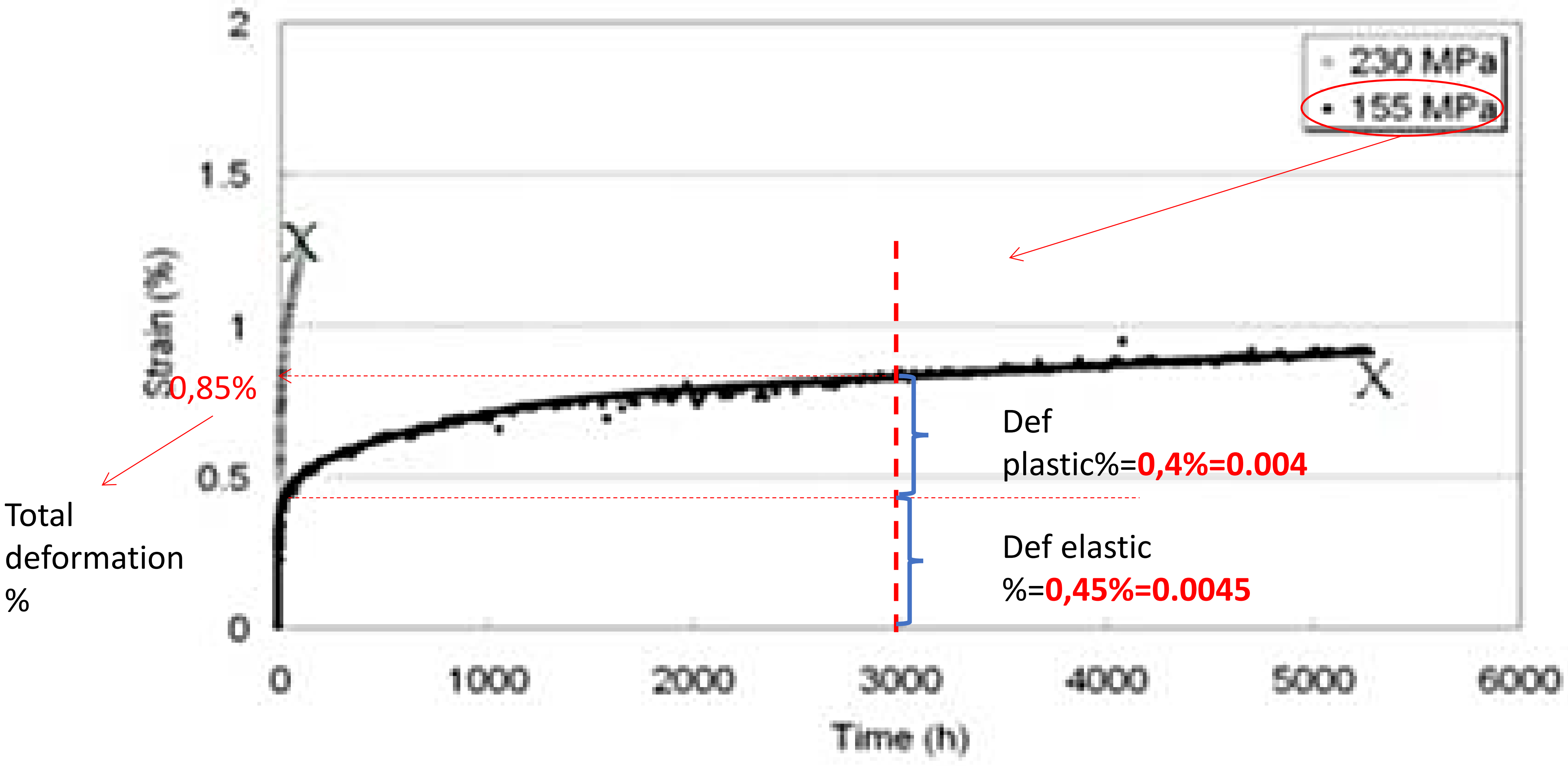
1.Instantaneous deformation increase → (you see by the stress strain curve:ex.4) given by the E decreases with temperature

Increasing temperature → more oscillations and distance between atoms
→<u>lower resistance of bonding strength to deformation</u>

3. Consequently Reduction time fracture

Ex.3 Indicate on the creep curve below the plastic and elastic deformation of a sample loaded at stress=155 MPa after 3000 hours.





230 MPa
155 MPa
Strain (%)
2
1.5
1
0,85%
0.5
0
0
1000
2000
3000
4000
5000
6000
Time (h)
Total deformation %
Def plastic%=0,4%=0.004
Def elastic %=0,45%=0.0045

Ex.4 Draw the stress/strain curves after tensile tests of a metal at room temperature (RT) and at T1 > RT. Briefly explain the difference.

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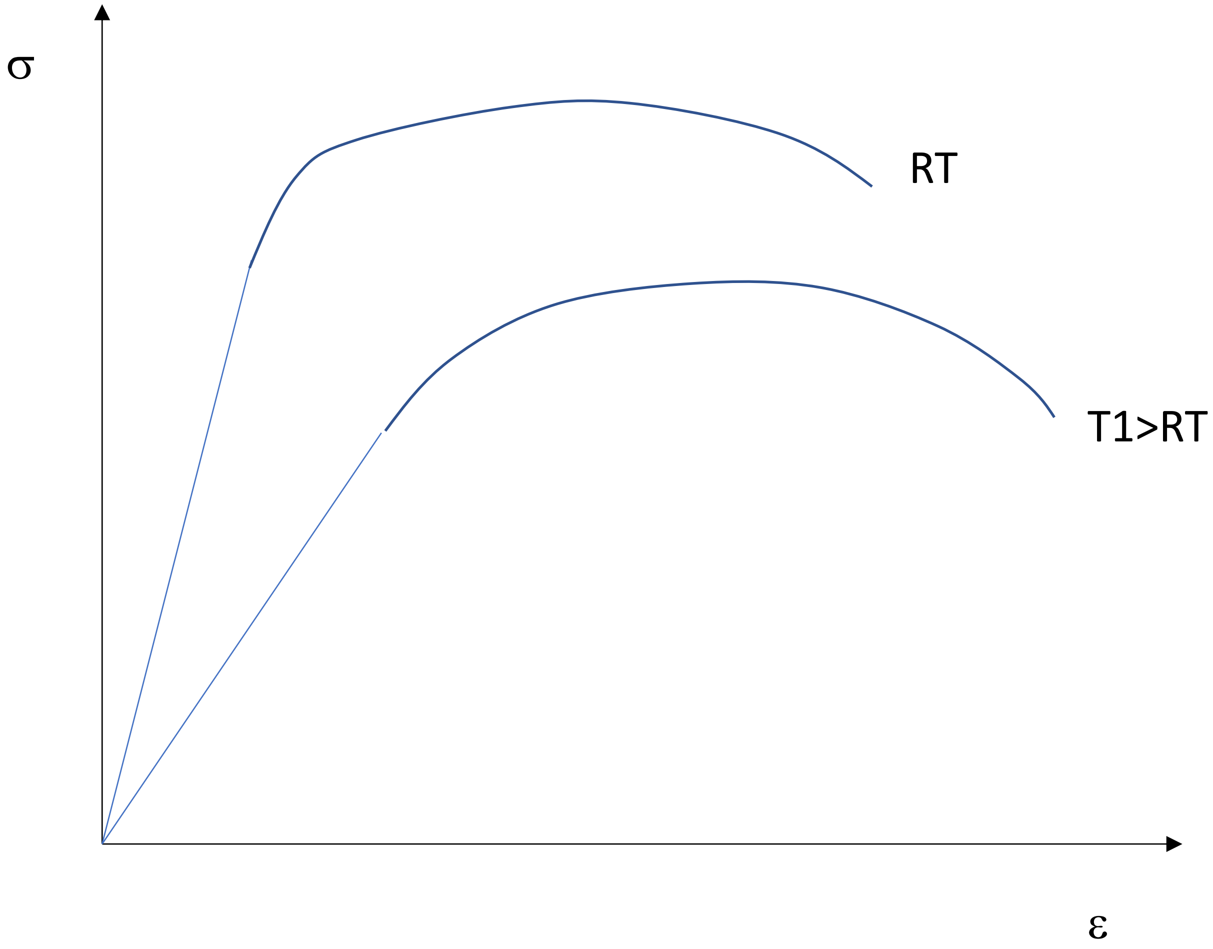
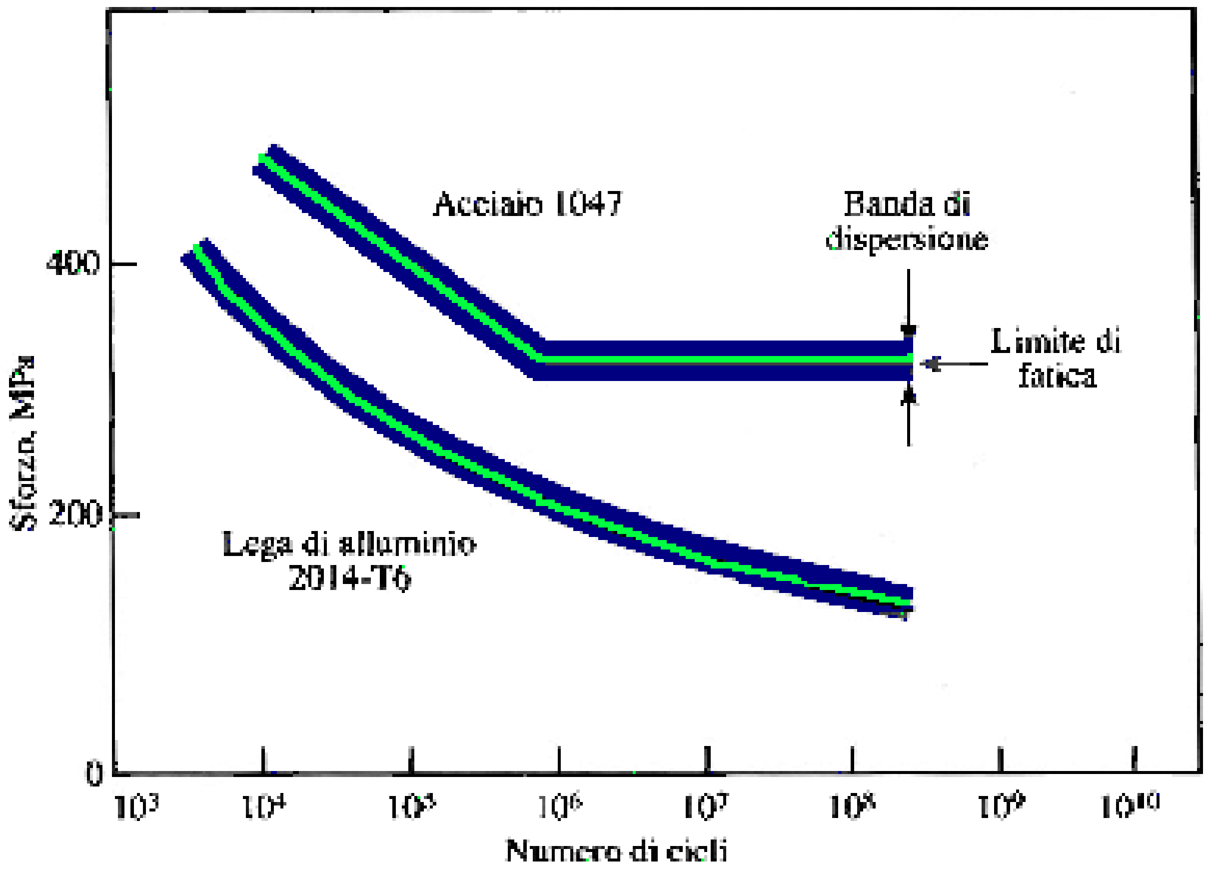


E decreases with temperature, increasing temperature oscillations of atoms increases as well as the distance between themselves

the interaction between atoms is lower (lower bonding strength → lower elastic modulus)

Fatigue

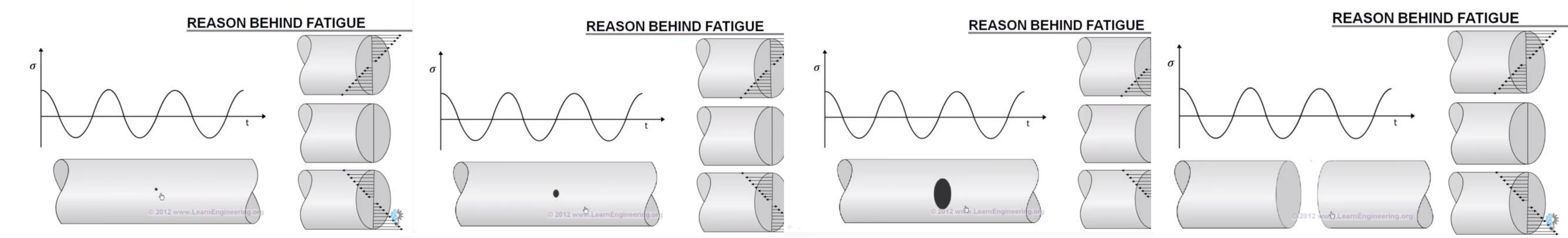
- Failure of materials due to cyclic loading.
- Main reason of mechanical failure of materials
- Failure happens at stress lower than σ_R or σ_Y
- Catastrophic failure of materials (also for ductile materials !)
- Fatigue tests: materials are cyclically loaded at different stresses up to failure.
- Fatigue limit: when cyclically loaded below this limit, materials do not fail (time considered by the graph)

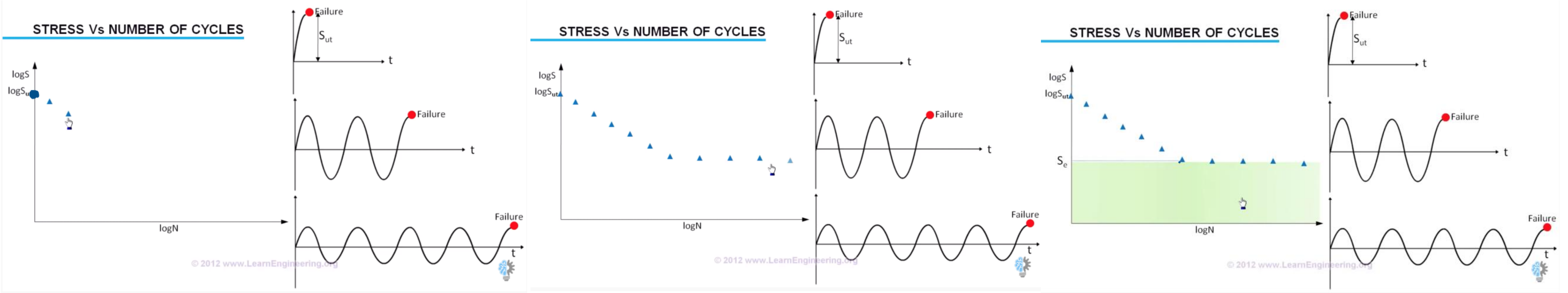


Fatigue

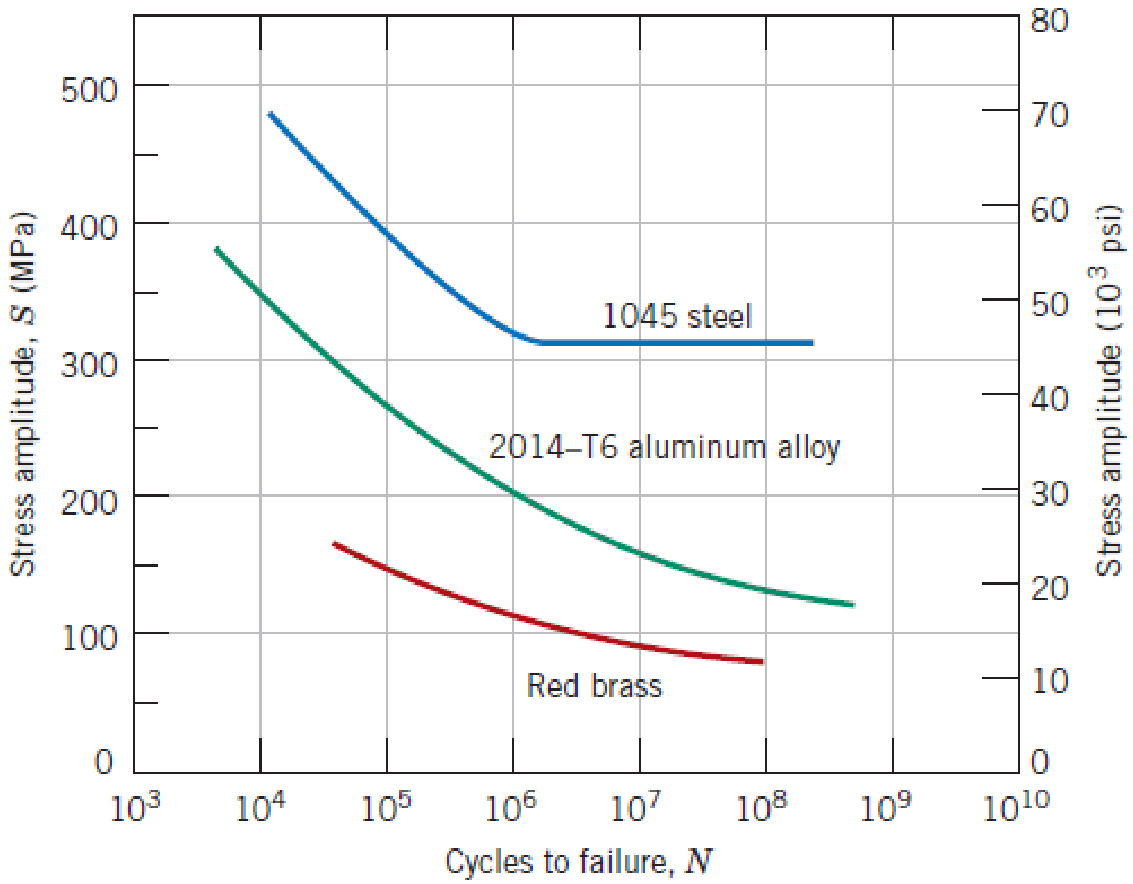
More cycles

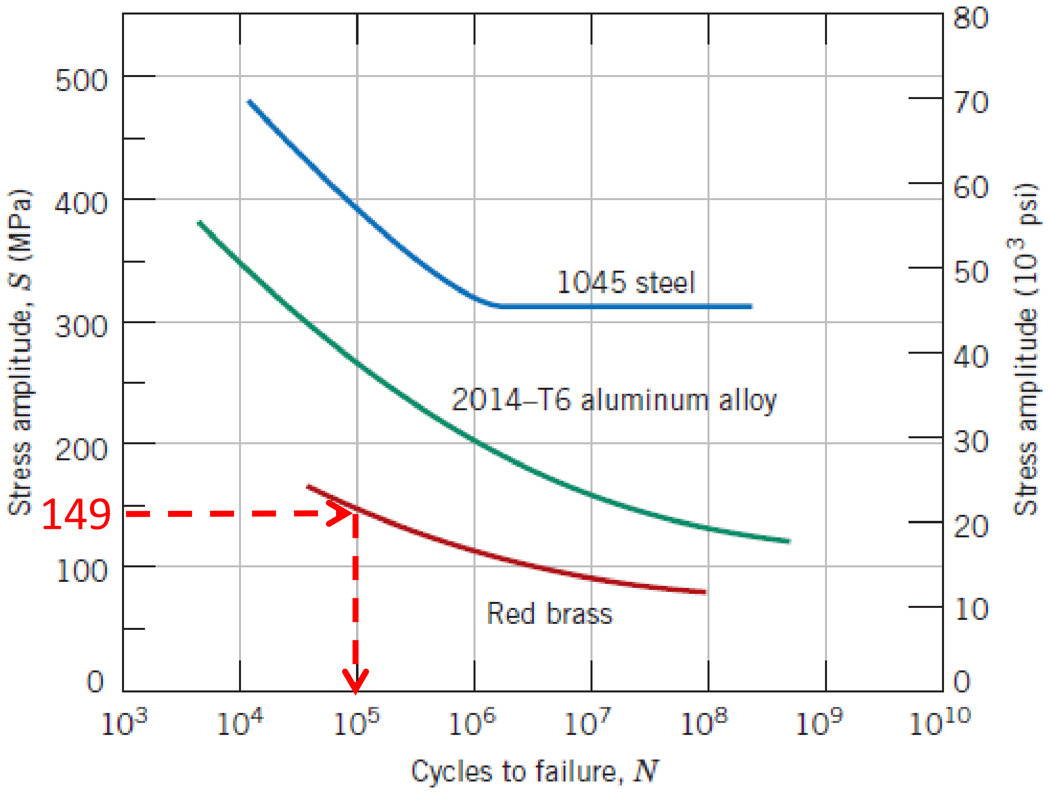
Oscillation about the constant amplitude stress→bring to the defect always higher





Ex.5 A Brass cylinder (8 mm diameter) is cyclically loaded at +7500 N e -7500 N along its axis. Use curves below to calculate its fatigue life.





$A = \pi r^2 = 3{,}14 * 16 = 50{,}24\ mm^2$

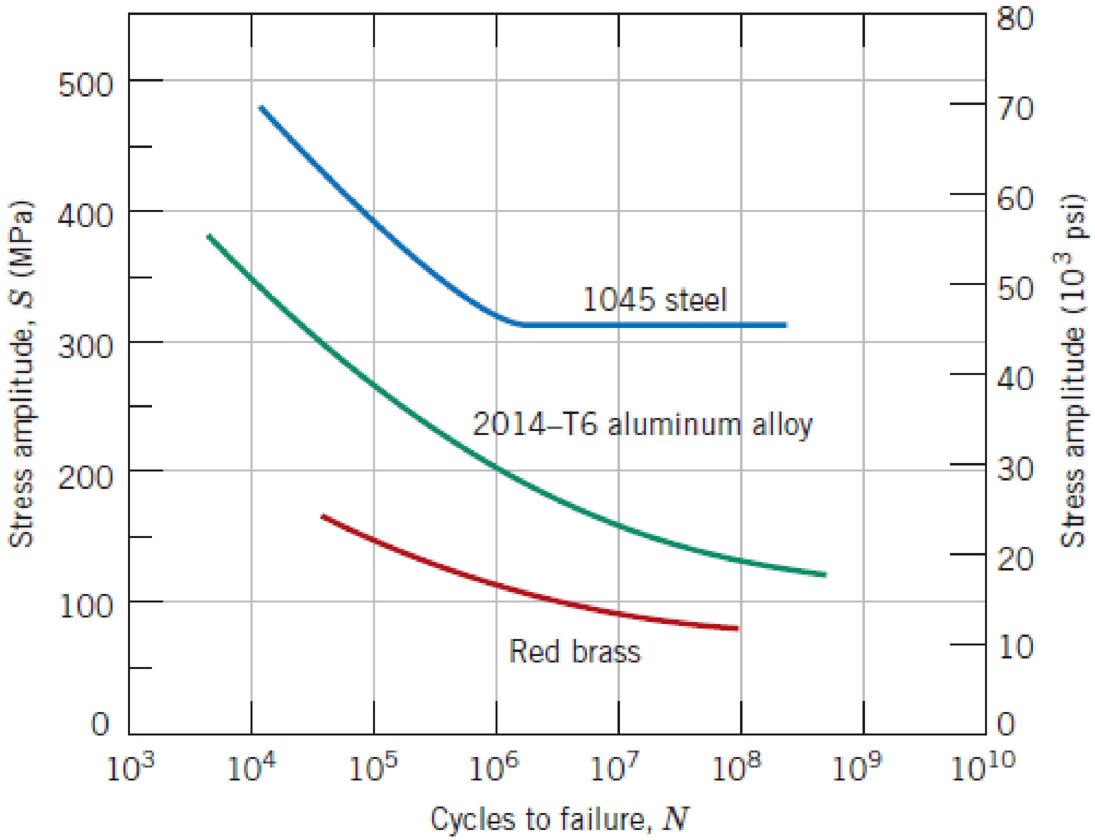
$\sigma_{max} = F_{max}/A = 7500\ N / 50{,}24\ mm^2 = 149\ MPa$

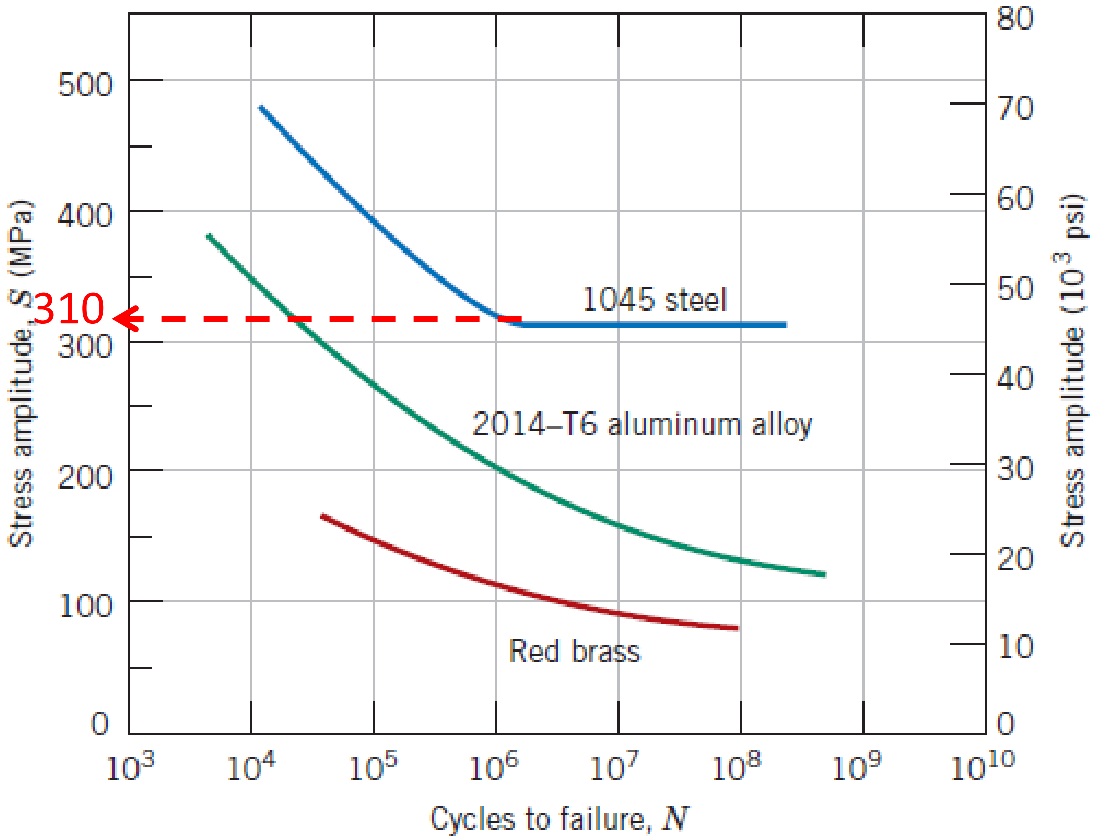
$\sigma_{min} = F/A = -\ 7500\ N / 50{,}24\ mm^2 = -149\ MPa$

$\sigma_m = (\sigma_{max} - \sigma_{min})/2 = 149\ MPa$

The fatigue life is **10^5** cycles

Ex.6 A steel rod is cyclically loaded at S= 22 kN along its axis. Use curves above to calculate the minimum diameter for this component to avoid fatigue fracture. Use a safety coefficient 2.





The fatigue limit of the steel is 310 MPa= σlim

To find minimum diameter:

σ=F/A

F=22kN=22000N

A=π (d/2)^2

σ=F/(π (**d**/2)^2)= 4F/ πd^2

F= σ / N = N safety factor=2

d=√(4F/(π(σ/N))=

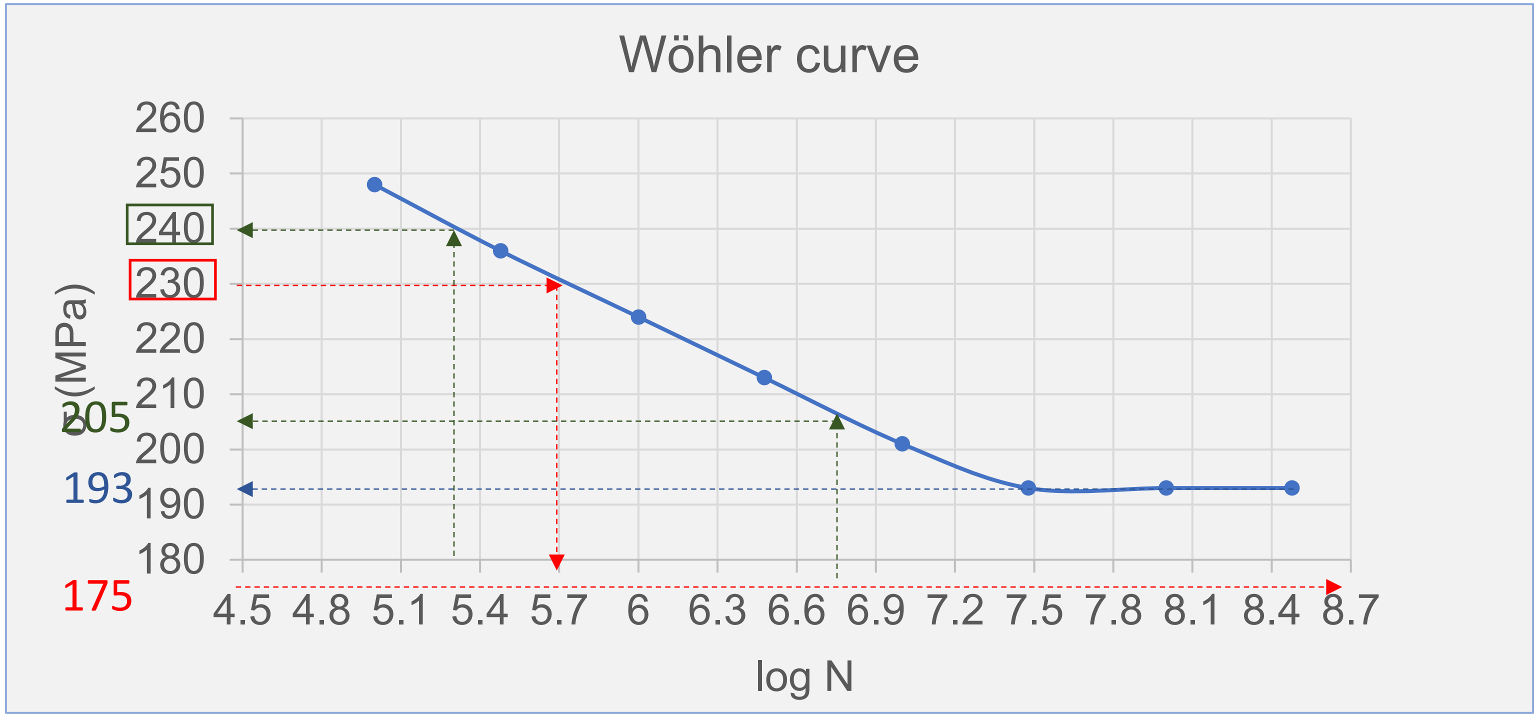
√ ((4*22000)/(π *(310 Mpa/2))) = **13** mm

Ex.7 The data obtained from a series of fatigue tests for a material are as follows:

σ (MPa)	N°
248	$1 \cdot 10^5$
236	$3 \cdot 10^5$
224	$1 \cdot 10^6$
213	$3 \cdot 10^6$
201	$1 \cdot 10^7$
193	$3 \cdot 10^7$
193	$1 \cdot 10^8$
193	$3 \cdot 10^8$

After constructing the Wöhler curve, determine:

a) The fatigue limit
b) The number of cycles to failure at 230 and 175 MPa
c) The fatigue strength at $2 \cdot 10^5$ and $6 \cdot 10^6$ cycles



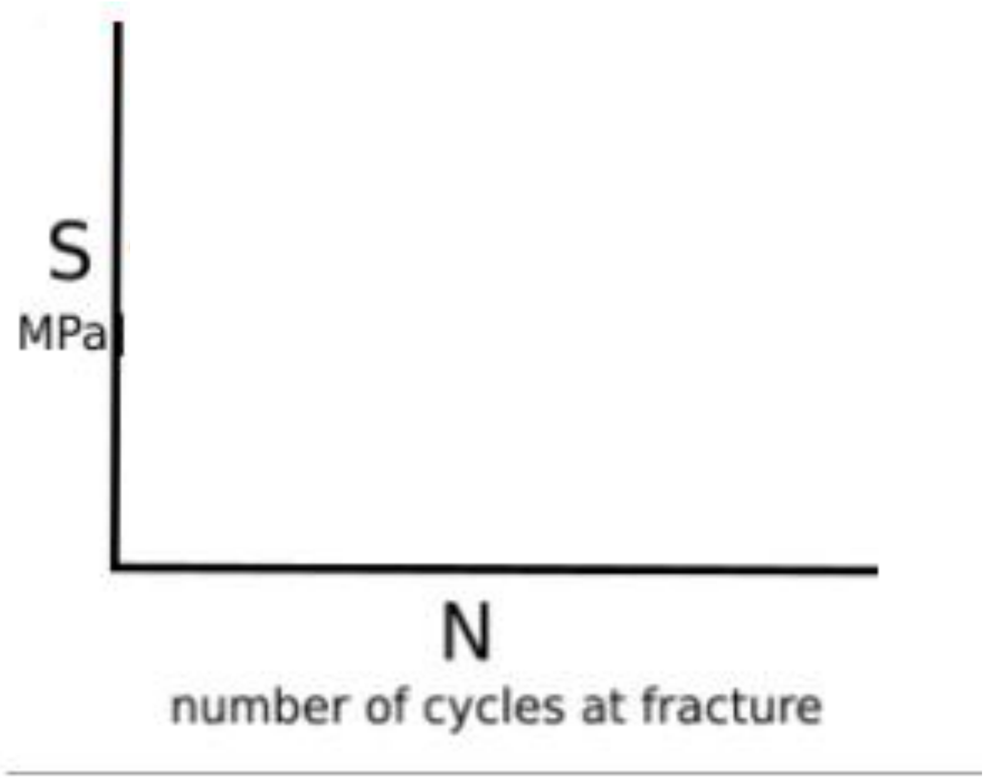
a) Fatigue limit : 193 MPa

b) Number of cycles to failure at 230 MPa: 501 187; $N=5*10^5$

Number of cycles to failure at 175 MPa: ∞

c) Fatigue strength at $2{\cdot}10^5$ cycles, $\log(2{\cdot}10^5)=5.3 \rightarrow 240$ MPa;
Fatigue strength at $6{\cdot}10^6$ cycles, $\log(6{\cdot}10^6)=6.8 \rightarrow 205$ MPa

Ex.8 Draw the curve (S/N) of a metal having the following surface characteristics: polished surface; sand-blasted surface; U-notches surface; V-notched surface. Explain differences.

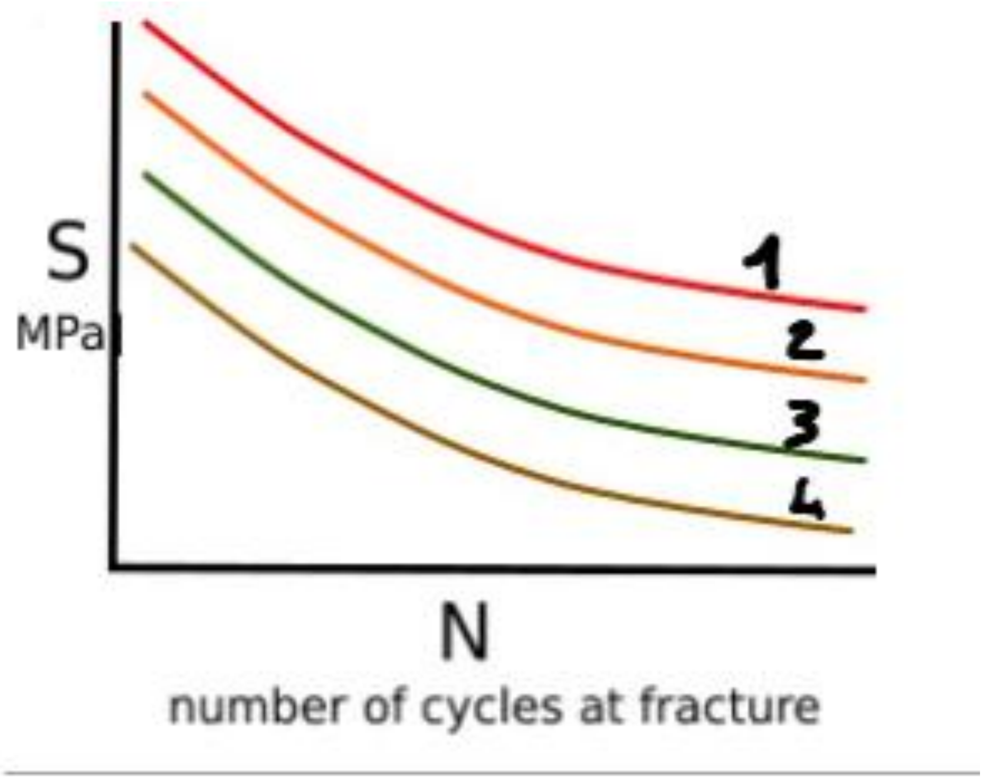


1- Polished

2- Sand-blasted

3- U-notch

4- V-notch



1- Polished→ smooth surface →no stress concentrators→ best fatigue strentgh

2- Sand-blasted→ Rough surface → small surface defects act as micro-notches

3- U-notch→rounded notch→lower fatigue strength but higher than V

4- V-notch, high concentration→ lowest fatigue strength